P.10 Inverse Functions

Name _____ Period ____ Date _____

Pre-Calculus P.10 · Student sheet

I can…

- verify that two functions are inverses.
- use the Horizontal Line Test to decide whether a function has an inverse function.
- find an inverse function algebraically.

Vocabulary · fill in the missing word

Inverse function — f^{-1}, where $f(f^{-1}(x)) =$ _____ $= x$

One-to-one function — a function in which each output comes from _____ one input

Horizontal Line Test — f has an inverse _____ if no horizontal line meets its graph more than once

Part 1 · Undo it

$f(x) = 3x - 5$. Fill in the table, then swap the columns.

x	0	1	2	3	4
f(x)					
Swapped: x					
Swapped: y					

Part 2 · Notes and examples (fill in each note, then work the example)

Verify: f and g are inverses if $f(g(x)) = x$ and _____ $= x$.

Example 1

Show that $f(x) = 2x^3 - 1$ and $g(x) = \sqrt[3]{\left(\frac{x+1}{2}\right)}$ are inverses.

Graphs: the graphs of f and f^{-1} are reflections in the line _____.

Example 2

f contains $(-2, -5)$, $(0, -1)$, and $(3, 5)$. Give three points on f^{-1}.

Horizontal Line Test: f has an inverse function if and only if no horizontal line meets the graph more than _____.

Example 3

Does each function have an inverse function?

a. $f(x) = x^3 + 1$

b. $g(x) = x^2 - 1$

Finding f^{-1}: replace f(x) with y, _____ x and y, and solve for y.

Example 4

Find the inverse of $f(x) = \frac{5 - 3x}{2}$.

Example 5

Find the inverse of $f(x) = \sqrt{2x - 3}$.

Part 3 · Partner practice

Partner 1

Find the inverse of $f(x) = 4x + 7$.

Partner 2

Find the inverse of $f(x) = x^3 - 2$.